

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Medium

Question

A circle in the xy -plane has the equation $(x - 13)^2 + (y - k)^2 = 64$. Which of the following gives the center of the circle and its radius?

Answer

- A. The center is at $(13, k)$ and the radius is 8.
- B. The center is at $(k, 13)$ and the radius is 8.
- C. The center is at $(k, 13)$ and the radius is 64.
- D. The center is at $(13, k)$ and the radius is 64.

Correct Answer: A

Rationale

Choice A is correct. For a circle in the xy -plane that has the equation $(x - h)^2 + (y - k)^2 = r^2$, where h , k , and r are constants, (h, k) is the center of the circle and the positive value of r is the radius of the circle. In the given equation, $h = 13$ and $r^2 = 64$. Taking the square root of each side of $r^2 = 64$ yields $r = \pm 8$. Therefore, the center of the circle is at $(13, k)$ and the radius is 8.

Choice B is incorrect. This gives the center and radius of a circle with equation $(x - k)^2 + (y - 13)^2 = 64$, not $(x - 13)^2 + (y - k)^2 = 64$.

Choice C is incorrect. This gives the center and radius of a circle with equation $(x - k)^2 + (y - 13)^2 = 4,096$, not $(x - 13)^2 + (y - k)^2 = 64$.

Choice D is incorrect. This gives the center and radius of a circle with equation $(x - 13)^2 + (y - k)^2 = 4,096$, not $(x - 13)^2 + (y - k)^2 = 64$.